

PBA: 15. Models of Uncertainty

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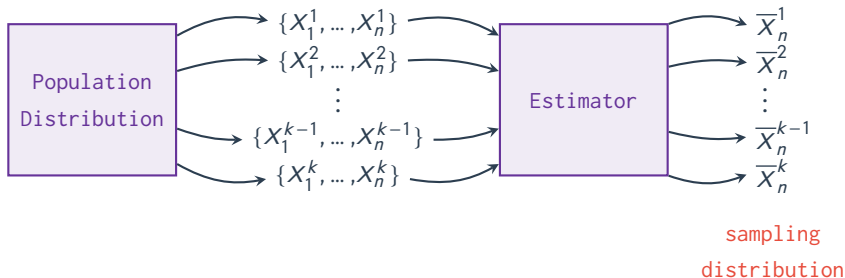
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Roadmap

1. Central Limit Theorem *Redux*
2. Using the Normal for Inference
3. Confidence Intervals for Experiments
4. Hypothesis Testing with the CLT
5. Two-Sample Tests

1/ Central Limit Theorem *Redux*

Sampling Distribution, in Pictures



- Sampling distribution: distribution of all possible values of a sample statistic computed using samples drawn from a given population.
 - Null distribution is a type of a sampling distribution created **under the setting where the null hypothesis is true**.
 - Bootstrap distribution $\rightsquigarrow$ approximation of the sampling distribution.

Sampling Distribution of the Sample Means

A sample mean (from a single random sample) can be 'decomposed' into the true pop. mean and chance error:

sample mean = population mean + chance error

$$\bar{X} = \mu + \text{chance error}$$

When examining multiple random samples, the sample means collectively form a sampling distribution, characterized by:

1. Central tendency: $\bar{X}$ centered at μ .
2. Spread: standard deviation of the sampling dist. is the **standard error**.

Spread of the Sample Means

- **Standard error:** how big is the chance error on average?
 - SD of the estimator **across repeated samples**.
 - With simple random samples, we can get a formula for the SE for any given estimator (e.g., sample mean).
- Standard error for the sample mean:

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{\text{population standard deviation}}{\sqrt{\text{sample size}}}$$

- Two components:
 1. Population SD (σ):
 - More spread out a variable is in the population $\rightsquigarrow$ more spread of sample means!
 2. Sample size (n):
 - larger sample $\rightsquigarrow$ spread of the sample means will be smaller!

Midwest Data Redux

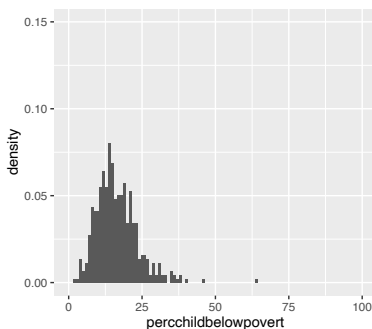
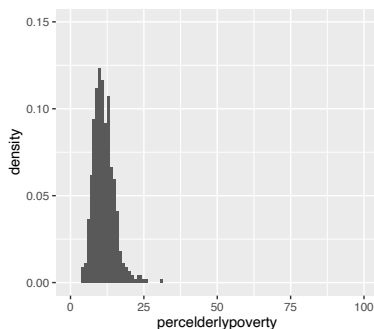
```
1 > library(ggplot2)
2 > midwest
3
4 # A tibble: 437 × 28
5   PID county state area poptotal popdensity popwhite popblack popamerindian popasian popother
6   <int> <chr> <chr> <dbl> <int> <dbl> <int> <int> <int> <int> <int>
7 1 561 ADAMS IL 0.052 66090 1271. 63917 1702 98 249 124
8 2 562 ALEXAN IL 0.014 10626 759 7054 3496 19 48 9
9 3 563 BOND IL 0.022 14991 681. 14477 429 35 16 34
10 4 564 BOONE IL 0.017 30806 1812. 29344 127 46 150 1139
11 5 565 BROWN IL 0.018 5836 324. 5264 547 14 5 6
12 6 566 BUREAU IL 0.05 35688 714. 35157 50 65 195 221
13 7 567 CALHOUN IL 0.017 5322 313. 5298 1 8 15 0
14 8 568 CARROLL IL 0.027 16805 622. 16519 111 30 61 84
15 9 569 CASS IL 0.024 13437 560. 13384 16 8 23 6
16 10 570 CHAMPA IL 0.058 173025 2983. 146506 16559 331 8033 1596
17 # 427 more rows
18 # 17 more variables: percwhite <dbl>, percblack <dbl>, percamerindian <dbl>, percasian <dbl>,
19 # percother <dbl>, popadults <int>, perched <dbl>, percollege <dbl>, percprof <dbl>,
20 # poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
21 # percchildbelowpovert <dbl>, percadultpoverty <dbl>, percelderlypoverty <dbl>, inmetro <int>,
22 # category <chr>
23 # Use `print(n = ...)` to see more rows
```

Data on socio-demographic info. for midwest counties from 2000 US **census**.

1. percelderlypoverty: Percent of elderly/seniors below poverty line.
2. percchildbelowpovert: Percent of children below poverty line.

Population Dists. in Midwest Countries

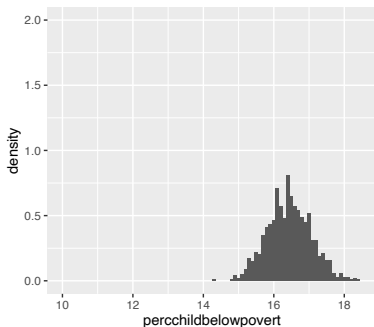
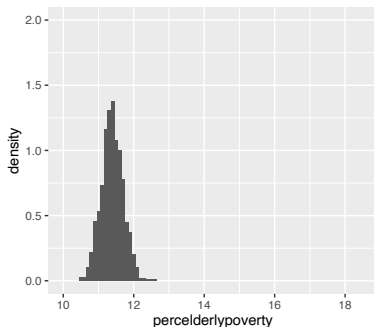
Population Distributions:



- Percent of elders/seniors in poverty (left) vs. Perc. of children in poverty (right):
 1. Which one has a higher spread?
 2. Are they normally distributed? Skewed?

Sampling Dists. from Midwest Countries

Sampling Distributions with $n = 100$ and $\text{reps} = 1,000$:

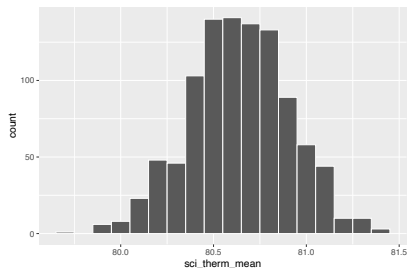
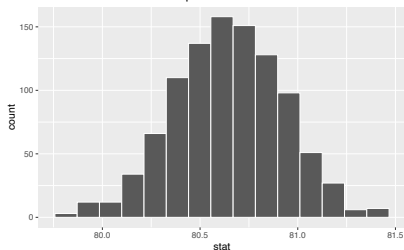


1. More population spread $\rightsquigarrow$ higher SE!
2. Are they normally distributed?

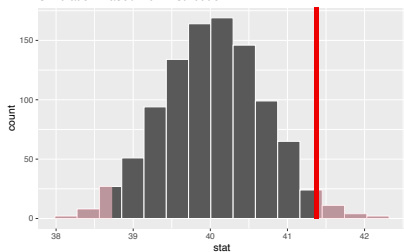
Similarity in the Bootstrap/Null Distributions

Therm. scores for scientists from the ANES data; working hours from the GSS data; ATEs from the social pressure experiment, etc. $\rightsquigarrow$ Use simulation-based methods to generate the dist.

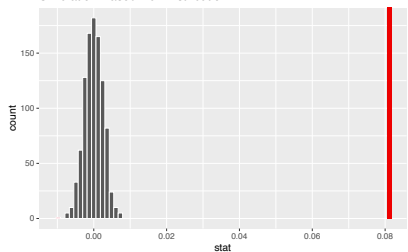
Simulation-Based Bootstrap Distribution



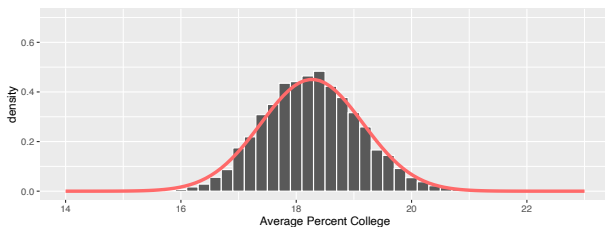
Simulation-Based Null Distribution



Simulation-Based Null Distribution



Conditions for the CLT

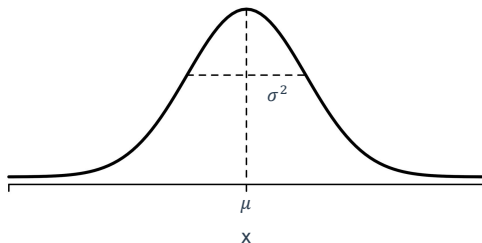


Central limit theorem: sums and means of 1) **simple random samples** tend to be normally distributed as the 2) **sample size grows**.

- CLT will still hold for non-simple random samples, e.g., stratified samples, w/in each stratum.

Key takeaway: Many, many **estimators will follow the CLT** and have a **normal distribution** $\rightsquigarrow$ easier to use this to make an inference as an alternative!

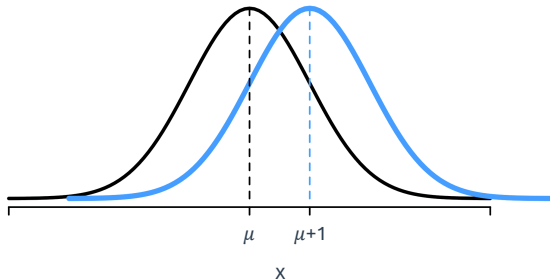
Normal Distribution



- A normal distribution can be affected by two values:
 - **Mean/expected value** usually written as μ
 - **Variance** written as σ^2 (standard deviation is σ)
 - Written $X \sim N(\mu, \sigma^2)$.
- **Standard normal distribution:** mean 0 and standard deviation 1.

Centering and Scaling the Normal

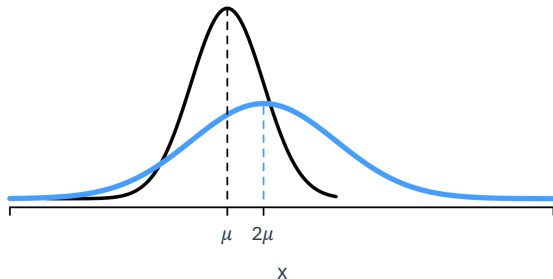
- Let $X \sim N(\mu, \sigma^2)$ and c be a constant.
- If $Z = X + c$, then $Z \sim N(\mu + c, \sigma^2)$.



- *Intuition?:* adding a constant to a normal shifts the distribution by that constant.

Centering and Scaling the Normal

- Let $X \sim N(\mu, \sigma^2)$ and c be a constant.
- If $Z = cX$, then $Z \sim N(c\mu, (c\sigma)^2)$.



- *Intuition?:* multiplying a normal by a constant scales the mean and the variance.

Z-scores of Normals

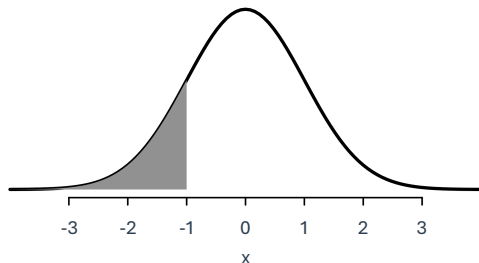
- These facts imply the **z-score** of a normal r.v. follows a std. normal:

$$z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

- If r.v. $X \sim N(\mu, \sigma^2)$
 - Centering: subtract the mean, $N(\mu - \mu, \sigma)$,
 - Scaling: divide by the SD, $N((\mu - \mu)/\sigma, (\sigma/\sigma)^2)$.
- z-score measures **how many SDs away from the mean a value of X is!**

Normal Probability Calculations

Q: What's the prob. of a std. r.v. x being below -1 in a standard normal?



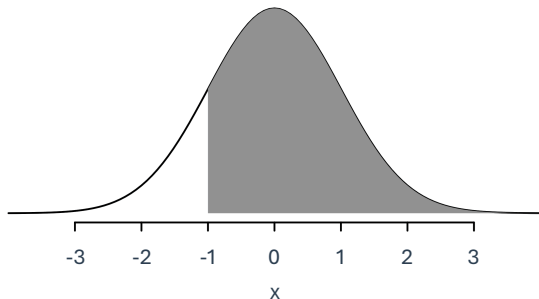
This is the area under the normal curve, which can be computed using `pnorm()`:

```
1 > pnorm(q = -1, mean = 0, sd = 1)
2 [1] 0.1586553
```

- Probability that an observation from a standard norm. dist. is less than or equal to q .

Normal Probability Calculations Cont.

Q: What's the probability of being **above** -1 in a standard normal?

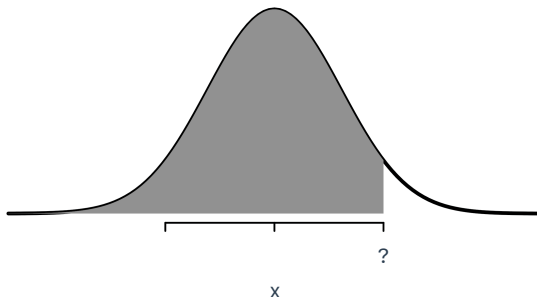


Total area under the curve (1) minus the area below -1 (i.e., prob. $q = -1$):

```
1 > 1 - pnorm(q = -1, mean = 0, sd = 1)
2 [1] 0.8413447
```

Normal Quantiles

Q: **What if we reverse the question?** what 'value' does x (w/ std. norm.) has to take to have 95% of the distribution to it's left?



This value is known as a **quantile** (cutoff point where a specified proportion of the distribution falls below it) which we can compute using `qnorm()`.

```
1 > qnorm(p = 0.95, mean = 0, sd = 1) |> round(digits = 2)
2 [1] 1.64
```

2/ Using the Normal for Inference

How Popular is Tesla's Cybertruck?



- What proportion of the market is interested in the Tesla Cybertruck?
- GLG market research poll/survey:
 - Polling period: late 2019
 - 43% are interested in purchasing, i.e., prospects (vs. 57% not interested)
- Define r.v. Y_i for Cybertruck demand:
 - $Y_i = 1 \rightsquigarrow$ respondent i interested in buying, 0 otherwise.
 - $p = \mathbb{P}(Y_i = 1)$ the population prop. of prospects interested in buying.
 - $\bar{Y} = 0.43$ is the sample proportion.

Standard Errors for Sample Proportions

- How 'variable' will our sample proportion be? $\rightsquigarrow$ Depends on the **standard error**.

- SEs of sample proportion, $\bar{Y}$:

$$SE \text{ for } \bar{Y} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{(\text{pop. prop.}) \times (1 - \text{pop. prop.})}{\text{sample size}}}$$

- Variance of a binary r.v. $\bar{Y}$: $p(1-p)$
- As p moves away from 0.5: more certainty that a randomly selected respondent will be interested in buying/not buying.
- Because we don't know p , we replace it with our best guess, $\bar{Y}$:

$$\widehat{SE} = \sqrt{\frac{\bar{Y}(1-\bar{Y})}{n}}$$

CLT for Confidence Intervals

- Given sample proportion, $\bar{Y}$, & population proportion, p :

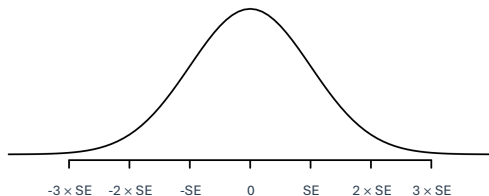
$$\bar{Y} - p = \text{chance error}$$

- How can we figure out a range of plausible chance errors?
 - So far with **bootstrap**, we used resampling to 'simulate' chance errors.
- Central limit theorem (CLT) implies:

$$\bar{Y} \approx N\left(p, \frac{p(1-p)}{n}\right)$$

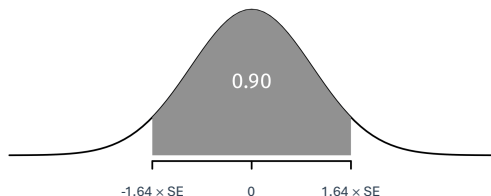
- Then, chance error ($\bar{Y} - p$) is approximately normal with mean 0 and SE equal to $\sqrt{p(1-p)/n}$.

Chance Errors



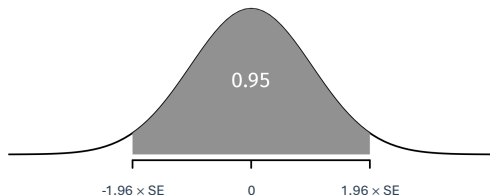
- If $\bar{Y} \sim N(p, SE^2)$, then the chance errors are $\bar{Y} - p \sim N(0, SE^2)$ so:

Chance Errors



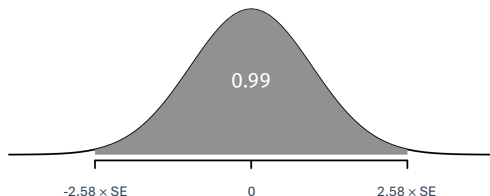
- If $\bar{Y} \sim N(p, SE^2)$, then the chance errors are $\bar{Y} - p \sim N(0, SE^2)$ so:
 1. $\approx 90\%$ of chance errors $\bar{Y} - p$ are within 1.64 SEs of the mean.

Chance Errors



- If $\bar{Y} \sim N(p, SE^2)$, then the chance errors are $\bar{Y} - p \sim N(0, SE^2)$ so:
 1. $\approx 90\%$ of chance errors $\bar{Y} - p$ are within 1.64 SEs of the mean.
 2. $\approx 95\%$ of chance errors $\bar{Y} - p$ are within 1.96 SEs of the mean.

Chance Errors



- If $\bar{Y} \sim N(p, SE^2)$, then the chance errors are $\bar{Y} - p \sim N(0, SE^2)$ so:
 1. $\approx 90\%$ of chance errors $\bar{Y} - p$ are within 1.64 SEs of the mean.
 2. $\approx 95\%$ of chance errors $\bar{Y} - p$ are within 1.96 SEs of the mean.
 3. $\approx 99\%$ of chance errors $\bar{Y} - p$ are within 2.58 SEs of the mean.

$\rightsquigarrow$ 1.64, 1.96, 2.58 are the quantile values for the corresponding normal probabilities (i.e., 90%, 95%, and 99%)
- This implies we can build a 95% CI with $\bar{Y} \pm 1.96 \times SE$ (color-coded in orange = margin of error).

3/ Confidence Intervals for Experiments

Comparison Between Groups

- So far: confidence intervals for means/averages.
- More interesting to compare across groups, e.g.,
 - Differences in public opinion across groups.
 - Difference between treatment and control groups.
- Bedrock of causal inference!

Social Pressure Experiment *Redux*

- Back to the Social Pressure Mail GOTV example for the primary election in Michigan 2006.
- Treatment group: postcards showing their own and their neighbors' voting records.
 - Sample size of treated group, $n_T = 360$ (artificially reducing the sample sizes for demonstration purposes)
- Control group: received nothing (no mail/postcards).
 - Sample size of the control group, $n_C = 1,890$

Outcomes of Interest

- Outcome: $Y_i = 1$ if a respondent i voted, 0 otherwise.
- Turnout rate (i.e., sample mean) in treated group, $\bar{Y}_T = 0.37$
- Turnout rate (i.e., sample mean) in control group, $\bar{Y}_C = 0.30$
- Estimated **average treatment effect (ATE)**:

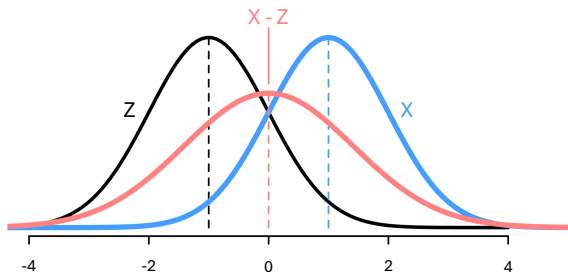
$$\widehat{ATE} = \bar{Y}_T - \bar{Y}_C = 0.07$$

Inference for the Difference

- Parameter: **population ATE** $\mu_T - \mu_C$
 - μ_T : Turnout rate in the population if everyone received treatment.
 - μ_C : Turnout rate in the population if everyone received control.
- Estimator: $\widehat{ATE} = \bar{Y}_T - \bar{Y}_C$
- By the CTL in large samples, we know that:
 - $\bar{Y}_T \approx N\left(\mu_T, \frac{\mu_T(1-\mu_T)}{n_T}\right)$
 - $\bar{Y}_C \approx N\left(\mu_C, \frac{\mu_C(1-\mu_C)}{n_C}\right)$
 - $\rightsquigarrow \bar{Y}_T - \bar{Y}_C \approx N(\mu_T - \mu_C, SE_{\text{diff}}^2)$
- Q: But, how do we compute SE_{diff} in this case?

Spread of a Difference in Normals

If we take the difference between two independent normal r.v.s, what happens to the spread?



⇒ The spread of the difference is **larger** than the spread of the two variables being differenced!

Standard Error for the Estimated ATE

- **Add** the SEs of each group to find the SE for a difference in means:

$$SE_{\text{diff}} = \sqrt{SE_T^2 + SE_C^2}$$

- Using what we know about SEs for binary outcomes/r.v.s:

$$SE_{\text{diff}} = \sqrt{\frac{\mu_T(1 - \mu_T)}{n_T} + \frac{\mu_C(1 - \mu_C)}{n_C}}$$

- We know that chance errors follows a normal:

- $\bar{Y}_T - \bar{Y}_C - (\mu_T - \mu_C) \approx N(0, SE_{\text{diff}}^2)$

- $\rightsquigarrow$ Construct a 95% CI with $\widehat{ATE} \pm 1.96 \times SE_{\text{diff}}$

Inference for Confidence Intervals

- Recall that we don't know μ_T and μ_C ! What to do?
 $\rightsquigarrow$ Plug in our sample proportions (i.e., $\bar{Y}_T, \bar{Y}_C$) to estimate the SE:

$$\begin{aligned}\widehat{SE}_{\text{diff}} &= \sqrt{\frac{\bar{Y}_T(1 - \bar{Y}_T)}{n_T} + \frac{\bar{Y}_C(1 - \bar{Y}_C)}{n_C}} \\ &= \sqrt{\frac{0.37 \times 0.63}{360} + \frac{0.30 \times 0.70}{1890}} = 0.028\end{aligned}$$

- We can now construct the confidence intervals based on CLT:

$$\begin{aligned}CI_{95} &= \widehat{ATE} \pm 1.96 \times \widehat{SE}_{\text{diff}} \\ &= 0.07 \pm 1.96 \times 0.028 \\ &= 0.07 \pm 0.054 \\ &= [0.016, 0.124]\end{aligned}$$

- Gives us the 'range of possibilities' taking into account plausible chance errors.

4/ Hypothesis Testing with the CLT

Statistical Hypothesis Testing

- Q: What would the world look like **if we knew the truth?**
 - That is, if we assume the null is true, what would our 'data' look like?
- Statistical hypothesis testing is a **thought experiment**.
 - Why 'thought' experiment? b/c we are essentially examining the what-if scenarios (i.e., what if the null were true?) when we formulate/test a hypothesis.
- Conducted with several steps:
 1. Specify your **null** and **alternative hypotheses**.
 2. Choose an appropriate **test statistic** and level of test α .
 3. Derive the **reference/null distribution** of the test statistic under the null.
 4. Use this distribution to calculate the **p-value**.
 5. Use p-value to **decide** whether to reject the null hypothesis or not.

How Popular is Tesla's Cybertruck?



- What proportion of the market is interested in the Tesla Cybertruck?
- Artificial/example market research poll: $\bar{Y} = 0.42$ with $n = 812$
- Can we reject the null that the population demand for Cybertruck is 50%?
 - Null?: $H_0 : p = 0.5$
 - Alternative?: $H_a : p \neq 0.5$

CLT for Hypothesis Testing

- Under the null ($p = 0.5$), we know the distribution of $\bar{Y}$:

$$\bar{Y} \approx N\left(p, \frac{p(1-p)}{n}\right) = N\left(0.5, \frac{0.5 \times 0.5}{812}\right)$$

- Using the **rules of normal transformations** where $X \sim N(\mu, \sigma^2)$:

$$\frac{X - \mu}{\sigma} \sim N(0, 1)$$

- Then under the null, we know the distribution of the test statistic (Z):

$$Z = \frac{\bar{Y} - 0.5}{0.5/\sqrt{812}} \approx N(0, 1)$$

Computing P-Values

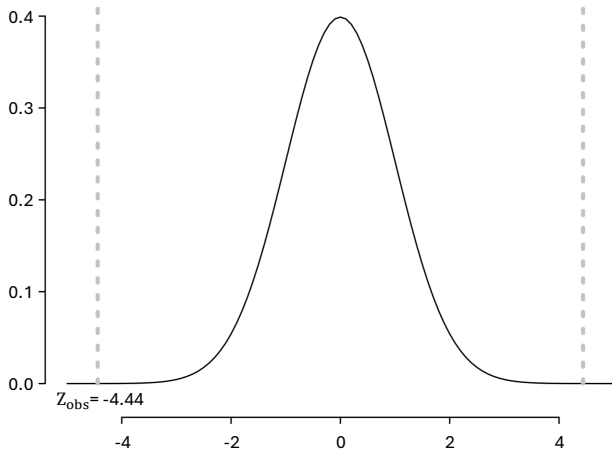
What we observe:

$$\begin{aligned} Z_{\text{obs}} &= \frac{\bar{Y} - 0.5}{0.5/\sqrt{812}} = \frac{0.42 - 0.5}{0.5/\sqrt{812}} \\ &= -\frac{0.08}{0.018} = -4.44 \end{aligned}$$

- Our observed sample proportion is 4.44 SEs away from 0.5 under the null.
- What is the **probability of being that far away?** $\rightsquigarrow$ **p-value!**

```
1 > options(scipen=999) # force R not to print numbers using sci. notations
2
3 > pnorm(q = -4.44, mean = 0, sd = 1) + # prob. being below -4.44
4   (1 - pnorm(q = 4.44, mean = 0, sd = 1)) # prob. being above 4.44
5
6 [1] 0.000008995888
```

- Q: What do we conclude about the demand for Cybertruck? $p = 0.5$?
 $p \neq 0.5$?



Generalizing Hypothesis Tests

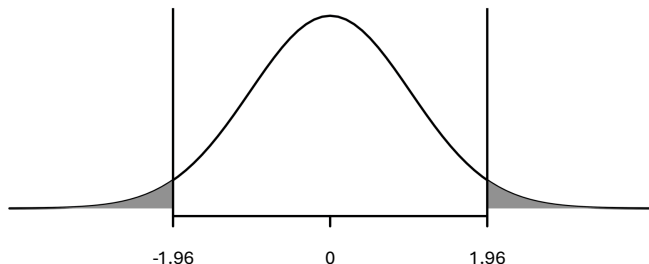
- Hypothesis testing using the CLT follows these general steps no matter what the estimator of our interest is.
 - Hypotheses: $H_0 : \mu = \mu_0$ (null guess); $H_1 : \mu \neq \mu_0$
 - Test statistic:

$$Z = \frac{\text{observed value} - \text{null guess}}{\widehat{SE}} = \frac{\bar{Y} - \mu_0}{\widehat{SE}}$$

↪ The exact estimator for the standard error, $\widehat{SE}$ will depend on the estimator of our interest.

- Null distribution: $Z \approx N(0, 1)$ by the CLT!
- P-value: probability of a standard normal being bigger than obs. test statistic (i.e., $|Z_{\text{obs}}|$).

Rejecting Regions



- We reject the null if p-value is below α (usually 0.05):
 - We know 5% of the time Z will be bigger (smaller) than 1.96 (-1.96).
 - That is, if $Z_{\text{obs}} > 1.96$ or $Z_{\text{obs}} < -1.96$, then the p-value must be below 0.05.
 - Thus, we **reject the null if $|Z_{\text{obs}}| > 1.96$** .

5/ Two-Sample Tests

Two-Sample Hypotheses

- Parameter: **population ATE** $\mu_T - \mu_C$
- Goal: learn about the population difference in means.
- Usual null hypothesis: no difference in population means (ATE = 0)
 - Null: $H_0 : \mu_T - \mu_C = 0$
 - Two-sided alternative: $H_1 : \mu_T - \mu_C \neq 0$
- Core question at trial in words?:
 - ↪ Q: are the differences in sample means just due to chance?

Difference-in-Means Review

- Sample turnout rates: $\bar{Y}_T = 0.37, \bar{Y}_C = 0.30$
- Sample sizes: $n_T = 360, n_C = 1890$
- Estimator of our interest? **sample difference-in-means:**

$$\widehat{ATE} = \bar{Y}_T - \bar{Y}_C = 0.07$$

- Estimated SE for the difference-in-means?:

$$\widehat{SE}_{\text{diff}} = \sqrt{\frac{\bar{Y}_T(1 - \bar{Y}_T)}{n_T} + \frac{\bar{Y}_C(1 - \bar{Y}_C)}{n_C}} = 0.028$$

CLT Again and Again

- Earlier, we saw that CLT gives us:

$$\bar{Y}_T - \bar{Y}_C \approx N(\mu_T - \mu_C, SE_{\text{diff}}^2)$$

- We can get the **test statistic** by standardizing the difference-in-means:

$$Z = \frac{(\bar{Y}_T - \bar{Y}_C) - (\mu_T - \mu_C)}{SE_{\text{diff}}} \sim N(0, 1)$$

- Same 'general form' of the test statistic as with one sample mean/proportion:

$$\frac{\text{observed} - \text{null guess}}{SE}$$

The Usual Null of No Difference

- Null hypothesis: $H_0 : \mu_T - \mu_C = 0$
- Test statistic:

$$Z = \frac{(\bar{Y}_T - \bar{Y}_C) - (\mu_T - \mu_C)}{SE_{\text{diff}}} = \frac{(\bar{Y}_T - \bar{Y}_C) - 0}{SE_{\text{diff}}}$$

- In large samples, we can replace the true SE with an estimate:

$$\widehat{SE}_{\text{diff}} = \sqrt{\widehat{SE}_T^2 + \widehat{SE}_C^2}$$

Calculating P-Values

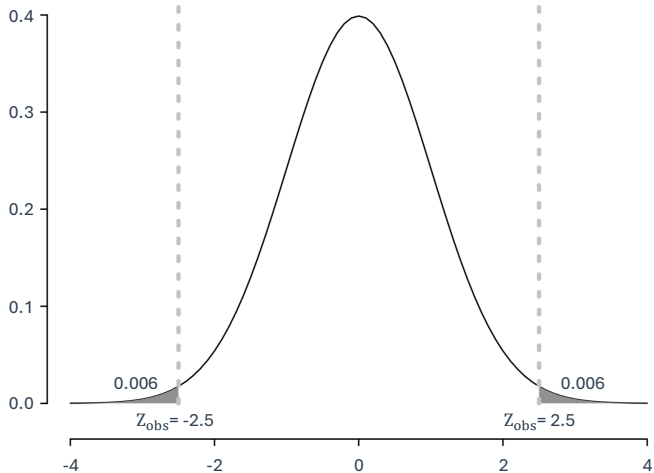
- Finally! Our test statistic in this sample:

$$Z = \frac{\bar{Y}_T - \bar{Y}_C}{\widehat{SE}_{\text{diff}}} = \frac{0.07}{0.028} = 2.5$$

- Calculate p-value given our test statistic (Z) of 2.5:

```
1 > pnorm(q = -2.5) + (1 - pnorm(q = 2.5))
2 [1] 0.01241933
3
4 > 2 * pnorm(2.5, lower.tail = FALSE) # another way to compute the two-sided p-value
5 [1] 0.01241933
```

- P-value based on a two-sided test: probability of observing a difference-in-means this big (or bigger) if the null were true.
 - Lower p-values $\rightsquigarrow$ stronger evidence against the null!
 - Q: **What do we conclude?**



Enjoy Your Weekends! :)

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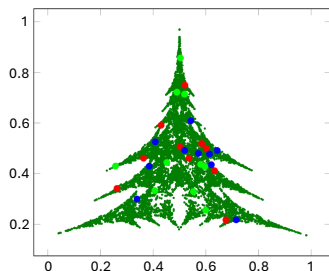
<https://j1yoo4.github.io/>



From last year, just before the festive season...

Merry Christmas!

= May your holidays be as merry as a perfectly plotted graph =



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